

AN INTERACTIVE JOB AND INTERNSHIP PLATFORM

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ABSTRACT

The Interactive Job and Internship Platform is a web-based system designed to connect job seekers, students, fresh graduates, and professionals with employers and companies

offering job and internship opportunities. This platform provides a centralized, efficient, and user-friendly interface where users can create profiles, upload resumes, search for jobs, and apply for positions. Employers, on the other hand, can post job openings, review applications, and communicate with potential candidates. One of the key features of the platform is its automated job matching system, which suggests relevant job openings based on a user's profile, skills, and interests. Employers also receive rec-

ommendations for suitable candidates, making the recruitment process faster and more efficient. The platform is developed using standard web technologies including HTML, CSS, JavaScript, and PHP, along with a database system for storing user and job-related information. The system aims to improve accessibility, reduce manual effort, and provide an efficient solution for job and internship searching. It aims to simplify and streamline the job search and hiring process, making it more accessible, time-saving, and effective for all stakeholders. In conclusion, this platform bridges the gap between job seekers and employers, providing an innovative and interactive solution to job and internship recruitment.

Keywords: Job and Internship Portal, Job Seekers, Employers, Automated Job Matching, Simple Profile Creation, Resume Upload, Job Search, Job Recommendations.

I. INTRODUCTION

In the current competitive employment landscape, a significant number of graduates face considerable difficulties during the transition from academic life to professional employment. Existing online platforms that aim to facilitate job searches

often fail to provide an inclusive and integrated environment where users can access opportunities across diverse sectors, such as private industries, government jobs, and overseas employment. Additionally, most platforms lack essential features such as personalized guidance, career counseling, and support services for internships and industrial training. The absence of these resources adversely impacts the preparedness and confidence of job seekers, particularly fresh graduates who often struggle with insufficient industry exposure and professional mentoring. To address these challenges, a unified platform is proposed which adopts a holistic approach towards job search and career development. It will include tools for resume building, interview preparation, and career counseling. Furthermore, dedicated sections for internships, industrial training, and mentorship programs will be implemented to assist users in gaining practical experience and professional insight. This platform is intended to act as a one-stop solution for students and graduates by broadening their employment reach and enhancing their career readiness. The inclusion of personalized services and continuously updated job listings ensures that users remain informed and connected in real-time with the job market.

II. LITERATURE REVIEW

The rapid growth of online recruitment platforms has significantly transformed the job search and hiring process. Various systems have been developed to improve accessibility, efficiency, and communication between job seekers and employers. This section reviews some of the existing works related to job portals and recruitment systems.

Mankawade A., et al., (IEEE 2023), Resume Analysis and Job Recommendation. A job recommendation system helps reduce the effort of job searching by analyzing resumes to identify suitable professions. Jobs from Naukri.com are matched based on the individual's skills and ranked for the user.

Mashayekhi Y., et al. (IEEE 2024)– Scalable Job Recommendation with Lower Congestion Using Optimal Transport. ReCon addresses congestion in job recommendations by using optimal transport and multi-objective optimization to fairly distribute vacancies among job seekers, while maintaining recommendation quality and scalability for large datasets.

Gupta G. K., et al. (IEEE 2021)– Enhancing Job Recommendations using Ensemble Learning Gupta et al. explore the application of ensemble learning techniques to enhance the accuracy of job recommendation systems. By combining multiple models, the study aims to improve the prediction of suitable job matches for candidates, addressing the challenges of diverse candidate profiles and job requirements.

III. PROPOSED SYSTEM

The proposed system is an interactive web-based platform designed to connect job seekers and employers through a simple and efficient interface. The system allows users to access job and internship opportunities without unnecessary complexity, focusing on ease of use and accessibility.

The platform enables users to create basic profiles and upload resumes, which can be used to apply for various job and internship opportunities. Employers can post job openings with relevant details such as job title, description, required skills, and location. This helps in organizing opportunities in a structured manner.

A key feature of the system is the job search functionality, where users can browse and filter available opportunities based on different criteria such as job type, domain, or location. This makes it easier for users to find suitable positions according to their preferences.

The system also supports direct interaction between job seekers and employers through application submission. The overall design ensures that the platform remains simple, efficient, and easy to navigate for all types of users.

The main functionalities of the system include:

- User profile creation and resume upload
- Job and internship posting by employers
- Job search and filtering based on user preferences
- Application submission to available opportunities

IV. SYSTEM ARCHITECTURE

The system architecture of the Job and Internship Platform is designed with a three-tier structure consisting of the User Interface Layer, Application Layer, and Database Layer, as shown in Fig. 1.

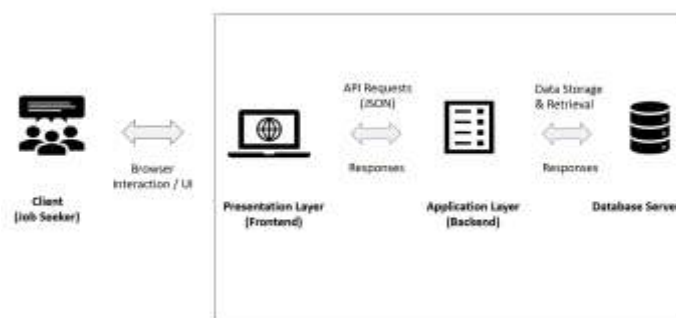


Fig. 1. System Architecture

The User Interface Layer allows interaction between users and the system through a web-based interface where job seekers can register, search, and apply for jobs, and employers can post and manage listings. The Application Layer acts as the intermediary, handling the business logic, validating user requests, and processing data between the user interface and the database. The Database Layer stores all essential information such as user details, job postings, and application records in a secure and organized manner.

V. METHODOLOGY

The proposed system follows a structured and modular approach for developing a web-based job and internship platform. The system is designed to provide a simple and efficient interaction between job seekers and employers through a centralized interface.

A. System Development Approach

The platform is developed using standard web technologies. The frontend is designed using HTML, CSS, and JavaScript to provide an interactive and user-friendly interface. The backend is implemented using PHP, which handles server-side processing and communication with the database. MySQL is used as the database to store user profiles, job listings, and application data.

B. User Profile Management

Users can create basic profiles by entering their personal details such as name, email, and skills. They can also upload their resumes, which are stored in the database and can be accessed by employers while reviewing applications.

C. Job Ranking Function

Jobs can be ranked using a weighted scoring formula:

$$R = w_1S + w_2E + w_3L \quad (1)$$

Where:

- S = Skill match score
- E = Experience match
- L = Location preference
- w_1, w_2, w_3 = weights

D. Job Posting Process

Employers can add job or internship opportunities by submitting details such as job title, description, required skills, and location. These details are stored in the database and made available for users to browse.

E. Job Search and Filtering

The system provides a search functionality that allows users to explore available job and internship opportunities. Users can filter results based on criteria such as category, location, or job type. This is achieved using database queries that retrieve relevant records based on user input.

F. Application Submission

Users can apply for jobs directly through the platform. When a user submits an application, their profile and resume information are sent to the respective employer and stored in the database for further processing.

G. System Workflow

The overall workflow of the system includes user interaction through the interface, processing of requests through the backend, and storage and retrieval of data from the database. This modular flow ensures smooth operation, scalability, and ease of maintenance.

The methodology ensures that the platform remains simple, efficient, and accessible while meeting the basic requirements of job seekers and employers.

VI. RESULTS AND DISCUSSION

The developed Job and Internship Platform was successfully implemented and tested in a controlled environment to evaluate its functionality and usability. The system demonstrates effective performance in connecting job seekers with employers through a simple and organized interface.

A.Feature Comparison

The comparison between the existing systems and the proposed system is shown in Table I. It highlights the improvements offered by the proposed platform.

TABLE I FEATURE COMPARISON OF JOB PLATFORMS

Features	Existing	Proposed
Profile Creation	Yes	Yes
Resume Upload	Yes	Yes
Job Search	Yes	Yes
Internship Support	Limited	Yes
User-Friendly	Moderate	High
Centralized	No	Yes

The table shows that the proposed system provides better usability, centralized access, and improved support for intern- ships compared to existing platforms.

B. Performance Analysis

The system performance was evaluated based on response time under different user loads, as shown in Fig. 2.

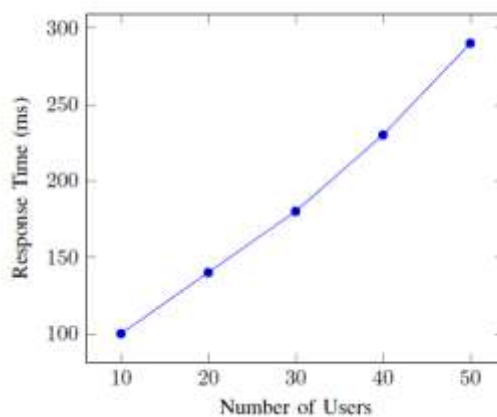


Fig. 2. System Performance Evaluation

The graph indicates that the response time increases gradu- ally with the number of users, but remains within acceptable limits. This demonstrates that the system performs efficiently under normal usage conditions.

C. Advantages of the System

- Provides a centralized platform for job and internship opportunities
- Reduces manual effort in job searching and recruitment
- Improves communication between job seekers and em- ployers
- Simple and easy-to-use interface for all users

D. Limitations

Although the system performs effectively, it has certain limitations:

- Does not include automated recommendation or matching features
- Limited to basic search and filtering functionalities
- Requires manual review of applications by employers Overall, the system achieves its goal of simplifying the job search and recruitment process while maintaining efficiency and usability. The results indicate that the platform can serve as a practical solution for connecting job seekers and employers in an organized manner.

VII. CONCLUSION AND FUTURE WORK

The proposed Job and Internship Platform provides a simple and effective solution for connecting job seekers and employ- ers through a centralized Web-based system. The platform successfully enables users to

create profiles, search for job and internship opportunities, and apply for positions in an organized manner. It also allows employers to post job openings and review candidate applications, thereby simplifying the overall recruitment process.

The system improves accessibility by providing an easy-to-use interface and reduces the time and effort required in traditional job searching methods. By integrating basic search and filtering features, the platform helps users find relevant opportunities more efficiently.

In future, the system can be enhanced by adding advanced features such as improved search mechanisms, better filtering options, and enhanced communication tools between users and employers. Additional improvements may include mobile compatibility, notification enhancements, and performance optimization to handle a larger number of users.

Overall, the platform serves as a practical and scalable solution for job and internship recruitment, with potential for further development and improvement.

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